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Fostering urban resilience and accessibility in cities: A dynamic knowledge graph approach

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ABSTRACT

This paper explores the utilisation of knowledge graphs and an agent-based implementation to enhance urban resilience and accessibility in city planning. We expand The World Avatar (TWA) dynamic knowledge graph to support decision-making in disaster response and urban planning. By employing a set of connected agents and integrating diverse data sources — including flood data, geospatial building information, land plots, and open-source data — through sets of ontologies, we demonstrate disaster response in a coastal town in the UK and various aspects relevant to city planning for a mid-sized town in Germany using TWA. In King's Lynn, our agent-based approach facilitates holistic disaster response by calculating optimal routes, avoiding flooded segments dynamically, assessing infrastructure accessibility before and during a flood using isochrones, identifying inaccessible population areas, guiding infrastructure restoration, and conducting critical path analysis. In Pirmasens, for city planning purposes, the knowledge graph-driven isochrone generation provides evidence-based insights into current amenity coverage and enables scenario planning for future amenities while adhering to land regulations. The implementation of agents and knowledge graphs achieves interoperability and enhances urban resilience and accessibility by enabling cross-domain correlation analysis that extends various areas including geospatial buildings, population demographics, accessibility coverage, and land use regulations.

1. Introduction

Global warming and climate change lead to severe weather disasters, giving rise to devastating events such as flooding (Poff, 2002; Schiermeier et al., 2011). Around 23% of the world population are directly exposed to 1-in-100-year floods (Rentschler, Salhab, & Jafino, 2022). These extreme weather disasters accelerate the need for cities to be highly adaptable and responsive to unexpected events. At the same time, other rare and unprecedented global crises like the COVID-19 pandemic have emphasised the importance of a city's hyper-locality attributes (Pisano, 2020). Due to transport lockdown and limitations in mobility, this sped up the development of 15-minute city initiatives. The 15-minute city is a concept which emphasises that all essential services such as healthcare, employment, cultural activities, and shopping should be conveniently located within a 15-minute radius from individuals' residences (Moreno, Allam, Chabaud, Gall, & Pratlong, 2021). These events greatly underscore the importance for cities to emphasise on agility and responsiveness to rapidly changing circumstances to attain holistic living conditions that are capable of fulfilling the residents' needs.

Cities have a wealth of information, yet they often face the hurdle of data silos and insufficient data integration. Many of the existing available data remains fragmented and untapped, *e.g.,* crowd-sourced data, building information, administrative land plots, *etc*. On top of this, current solutions in disaster response and city planning do not leverage on these data enough in their analysis. This unrealised potential in data is hindering the effectiveness of city planning and disaster response.

Effective decision making in disaster scenarios requires overcoming challenges related to the integration and the interoperability of heterogeneous datasets (Li, Wang, et al., 2023). Many severe weather solutions primarily concentrate on disaster response; however, these

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Nomenclature

API	Application Programming Interface
GeoSPARQL	Geographic Query Language for RDF Data
GML	Geography Markup Language
KG	Knowledge Graph
OBDA	Ontology-Based Data Access
OSM	OpenStreetMap
OS	Ordnance Survey
POI	Point of Interest
RDF	Resource Description Framework
SPARQL	SPARQL Protocol and RDF Query Language
SQL	Structured Query Language
TWA	The World Avatar

solutions fall short in providing capabilities beyond geospatial analysis (Janowicz et al., 2022). This situation can often result in the lack of information transfer between knowledge models which leads to poor scalability and data re-usability in systems, therefore hindering the development of a comprehensive disaster response strategy.

In 15-minute city planning, planning tools for city accessibility exhibit similar challenges in data integration and model re-usability. While there are models and applications developed by researchers that are effective in analysing 15-minute accessibility in cities, however their applicability is often limited to major cities (Nicoletti, Sirenko, & Verma, 2022) or limited to certain countries (Here Technologies, 2020). The limited attention given to smaller towns, often coupled with the lack of resources for small towns to engage premium urban planning analysis services pose challenges to the even distribution of essential amenities (Knox & Mayer, 2013). This can further accentuate inequality patterns in terms of social and transport mobility in economically disadvantaged or smaller cities (Guzman, Arellana, Oviedo, & Aristizábal, 2021).

One commonality across both challenges in disaster planning and urban planning is a high barrier of entry caused by the integration of diverse data sources. These data sources commonly come in its own formats, leading to compatibility issues and high set up time. The datasets often comprise a mix of geospatial data and tabular data, covering information about land plots, census data and building information, that can be very different in nature. This gap in data integration creates difficulties and complexities in achieving interoperability, further hindering the full potential of exploiting these data sets (Ding et al., 2023). This scenario could lead to researchers producing one-off location and use-case-specific analyses that are difficult to scale and transfer.

The integration of various datasets poses significant difficulties, however both cases of disaster response and city planning necessitate utilising these dataset for cross-domain analysis. A knowledge graph represents individual information in a structured graphical form, capturing relationships between entities (Sheth, Padhee, & Gyrard, 2019). Its unique interoperability and cross-domain functionality solves many constraints in integrating various data types to derive valuable insights, making it an excellent candidate in tackling the challenges in resilience planning and urban planning solutions.

The purpose of this paper is to address the challenge of integrating population, isochrone, flood raster, and building information open-source data for resilience and urban planning. It explores the potential of applying knowledge graphs in representing this information semantically to overcome this challenge. An agent-based approach is used to leverage the instantiated information on the knowledge graph. This enables disaster response capabilities such as routing avoiding floods, healthcare accessibility analysis during floods using isochrone, infrastructure restoration guidance, and critical path analysis. Similarly, the same agent-based approach is applied to city planning by using knowledge-graph driven isochrone generation to provide evidence-based insights into current amenity coverage and enable scenario planning for future amenities while adhering to land regulations. The implementation of the agent-based approach with knowledge graphs aims to enable cross-domain analysis that covers areas including geospatial buildings, population demographics, accessibility coverage, and land use regulations.

The sections of this paper are organised as follows: Section 2 details the problem and prior technical efforts; Section 3 introduces the methodology used and relevant data sources; Section 4 highlights the use cases and results; and Section 5 concludes the work.

2. Background

2.1. Urban resilience

In view of current climate change, disasters have become increasingly unpredictable (Hultman & Bozmoski, 2006). Cities must have the capacity to not only withstand and recover from disturbances, while also being able to plan and adapt to the constantly evolving circumstances (Coaffee & Lee, 2016; Sharifi, 2019). The frequency of extreme weather events poses a substantial threat to the daily lives of individuals. This situation highlights the need for government authorities to possess the right tools and resources to act promptly and effectively during life threatening situations. During extreme weather events, such as flooding, proper emergency route planning to rescue and evacuate civilians under distress can prevent the loss of life (Kolen, Kok, Helsloot, & Maaskant, 2013). Hence, adapting a strategy that incorporates multiple sources of data such as flood, road networks, population and emergency services locations would enhance effective resource allocation and strategic planning.

During flooding events, motorists often attempt to pass through flooded roads because they are unaware of alternative routes, which can be risky and fatal (Coles, 2020). This situation can be problematic for essential vehicles (*i.e.,* ambulance, rescue trucks) which have to navigate through flooded areas swiftly and safely. The current research done on flood-avoiding route planning does not use flood depth estimates in their assessments (Choosumrong, Humhong, Raghavan, & Fenoy, 2019; Singh et al., 2015; Veerubhotla, Bothale, Kumar, Urkude, & Shukla, 2015), thereby neglecting a crucial parameter that could potentially enhance the safety and effectiveness of route planning.

Other than that, some current disaster response strategies struggle with insufficient data and lack of data interconnectivity. For example, existing GIS-based emergency evacuation frameworks planning face challenges in assessing evacuation route planning during floods due to insufficient data (Parajuli, Neupane, Kunwar, Adhikari, & Acharya, 2023). One notable issue is the difficulty in accurately determining the locations of shelter houses to compute accessibility. Additionally, other studies on accessibility to critical amenities during flooding aims to enhance their comprehensiveness in disaster strategy solutions by integrating population distribution and evacuation routing capability (Alabbad, Mount, Campbell, & Demir, 2021). Through literature review, these disaster strategies each exhibit deficiencies in certain aspects such as insufficient data or functions. Therefore, there exists a research gap to integrate flood depth data, buildings location, and population distribution to build a comprehensive, well-informed disaster response strategy.

2.2. The 15-minute city

15-minute city is an urban planning concept developed by Carlos Moreno (Moreno et al., 2021) which describes that residents should be able to satisfy their daily needs which include work, home, food, health, education, culture, sports, and leisure within 15 min of walking or cycling from their residence. COVID-19 pandemic has prompted a widespread shift towards the concept of 15-minutes cities as cities globally adapt in response to lockdown restrictions (Li, Chai, Chen, & Li, 2023). The constraints on mobility resulted in the increasing tendency to centralise essential amenities in close proximity. The primary objective of this movement is to ensure that all residents can conveniently access all these necessary services, emphasising equal access to urban services for all people (Khavarian-Garmsir, Sharifi, & Sadeghi, 2023).

Across the globe, governments are strategically pushing for proximity-based planning initiatives (Papadopoulos, Sdoukopoulos, & Politis, 2023). Notably, in Paris, the city's mayor Anne Hidalgo envisions the “Ville Du Quart D'heure" (Quarter-hour city) (Pozoukidou & Chatziyiannaki, 2021). Singapore has introduced the "20 Minute Town, 45 Minute City Master Plan" (Manifesty & Park, 2022), while Melbourne's Victoria government is committed to the concept of "20-Minute Neighbourhood" (State Government of Victoria, 2023). In Shanghai, local district authorities are advocating the idea of a “15-Minute Town" (Information Office Of Shanghai Municipality, 2023). Despite variations in the specified time frames in these concepts, the common thread across these strategies is the focus on proximity-based planning and reduction on car reliance (Moreno et al., 2021).

Gaining widespread attention, 15-minute city has also become a popular topic among researchers. The 20 min city analysis was performed in Liverpool from a socio-spatial inequalities perspective (Calafiore, Dunning, Nurse, & Singleton, 2022). The X-Minute City analysed the 500 largest cities in the United States and 43 urban areas of New Zealand in the intervals of 10, 15, 20 minutes (Logan et al., 2022). The 20-Minute Neighbourhoods in Toronto assessed the walk-ability of neighbourhoods (Sulaiman & Abulajiang, 2023). Similar qualitative analyses and spatial evaluations were conducted for the 15-minute city model in Oslo and Lisbon (Di Marino, Tomaz, Henriques, & Chavoshi, 2023).

The adoption of proximity-based planning can aid policymakers in strategically planning and designing liveable and densified cities that are aligned with the UN Sustainable Development Goals to create inclusive, safe, resilient, and sustainable cities (United Nations, 2015). Not only that, the importance of accessibility extends beyond mere convenience, it plays a crucial role in fostering a more equitable and thriving society, as having higher accessibility and better walkability in a city often leads to a higher quality of urban life (Weng et al., 2019).

However, as observed across the current literature, researchers often conduct location-specific studies that may become outdated as cities and amenities evolve. While there are online applications to assess accessibility by generating isochrones - a polygon that describes the area of reach within a specified time frame, such as ‘Iso4app‘(K-SOL S.r.l, 2023), ‘Geoapify‘(Geoapify GmbH, 2023) and ‘TravelTime‘(TravelTime, 2023), these existing off-the-shelf solutions do not include the existing building infrastructure in their analyses. To retrieve and integrate the results of the generated isochrones, these services come with pricing plans that allow users to request by API calls specifying accurate locations. For institutions handling sensitive data, such an approach might be unsuitable without disclosing sensitive information externally. Other more comprehensive 15-min city planning apps done by researchers, such as CityAccessMap (Nicoletti et al., 2022) and 15-Min-City-App (Here Technologies, 2020) focus only on major cities and typically these studies risk becoming obsolete over time as the urban landscape changes.

A research study commissioned by the European Union to evaluate the effectiveness of various accessibility instruments concluded that while isochrones are excellent instruments to identify low accessibility areas, there is a notable challenge on the integration of diverse data sources from various sources such as land use (Te Brömmelstroet et al., 2014). Other urban planning tools that generate isochrones in Germany have been assessed and found to exhibit limitations, particularly in their location-specific focus (Hull, Silva, & Bertolini, 2012). These tools face challenges when attempting to scale to measure specific indicators (*i.e.,* accessibility to certain amenities) in diverse regions (Hull et al., 2012). These scenarios underscore the need for an enhanced methodology to create a versatile tool adaptable to the dynamic needs of a city, operates independently of location, and is capable of seamlessly incorporating diverse data into analyses.

2.3. The World Avatar

The World Avatar (TWA) project (Akroyd, Mosbach, Bhave, & Kraft, 2021) objective is to construct a digital ‘avatar’ of the world. The core concept behind TWA revolves around building an all-encompassing world model to facilitate interoperability between previously isolated yet conceptually connected domains, encompassing both knowledge and data. TWA relies on semantic web technologies and employs a dynamic knowledge graph to semantically describe relationships between concepts and instances. The term ‘dynamic’ signifies that TWA integrates ontologies (*i.e.,* data definitions) with the actual data instances (*i.e.,* from API), and also the computational services (*i.e.,* agents) that operate on the instantiated data (Hofmeister, Brownbridge, et al., 2023). These computational agents instantiate (i.e., create and modify) and derive (i.e., analyse and infer) information within the knowledge graph. The semantic agents constitute an integral part of the knowledge graph system, enabling semantic reasoning, therefore making TWA inherently dynamic.

TWA is modular and scalable by design, and new concepts and relationships can be added continuously while maintaining connections to everything existing. It allows decentralisation and enables interoperability across heterogeneous data sources and software. The intelligent agents that are part of TWA act as executable knowledge components, keeping the system current in time and self-evolving. "Agent" commonly refers to software, methods, applications, services, *etc*., that utilise semantic web technologies and operate on the knowledge graph to read, write, estimate, simulate, optimise, query, to fulfil specific objectives (Lim, Wang, Inderwildi, & Kraft, 2022). These agents communicate with one another via KG to derive insights (Eibeck, Lim, & Kraft, 2019). Information is exchanged via the KG to ensure an aligned world view among actors: input agents instantiate information into KG, while other agents use the instantiated data to perform tasks. Using this technology stack, TWA is versatile in three aspects: (1) answering cross-domain questions about the current world, (2) controlling real-world entities, and (3) supporting what-if scenario analysis. TWA follows a system of systems approach where individual task-oriented agents and their interplay help to describe complex behaviour to gain a comprehensive understanding and foster informed decisions.

Ontologies serve as structured representations of knowledge, encapturing concepts, relationships, and properties within a specific domain. They establish a formal framework for organising and sharing information in a machine-readable format. The representation of data through ontologies results in the formation of directed graphs, known as knowledge graphs (KGs), where nodes define entities and data (*i.e.,* concepts or instances) and edges denote their relationships (Akroyd et al., 2021). KGs offer an extensible data structure that is well suited for representing arbitrarily structured data and which can be hosted decentralised, *i.e.,* distributed over the internet, using Semantic Web technology (Berners-Lee, Hendler, & Lassila, 2001). This implementation is also known as Linked Data (Bizer, Heath, & Berners-Lee, 2011), where every concept and relation is reference-able to its definition, facilitating information discovery across the web and enhancing machine readability and automation.

Linked Data uses resource description framework (RDF) (Klyne, 2004) as the standard language to store data in the form of subject, predicate, and object triples (Akroyd et al., 2021). Subject represents a resource, the predicate indicates a specific property and the object represents a value or another resource (Hofmeister, Brownbridge, et al., 2023). For example in the context of isochrone, concepts like `Isochrone`, `TransportMode`, `Buildings` can be defined in an ontology. While Linked Data connect these concepts through subject–predicate–object triples, such as `Isochrone A - assumesTransportMode - Car` or `Isochrone B - originatesFrom - Building X`.

The ontologies applied in this work include OntoFlood (Hofmeister, Bai, et al., 2024), OntoBuiltEnv (Hofmeister, Bai, et al., 2024), OntoCityGML (Chadzynski et al., 2021), OpenGIS ontology (Battle & Kolas, 2011). OntoFlood organises flood-related data and its disaster impact (Hofmeister, Bai, et al., 2024). While OntoBuiltEnv organises building related metadata ranging from its building usages, number of rooms, energy ratings, property prices, construction dates, *etc*. Hofmeister, Bai, et al. (2024). OntoCityGML provides a knowledge graph representation to the geometric representation of urban objects such as building heights, roof type, floor surface, wall surface, material *etc*. Chadzynski et al. (2021). Additionally, the OpenGIS ontology (Battle & Kolas, 2011) is used to provide structured representation of geospatial concepts.

These ontologies are interconnected in several ways, for instance, OntoBuiltEnv and OntoCityGML are directly interconnected through a common ontology class of `obe:Property` that represents building. While OntoCityGML and OntoFlood both use the OpenGIS ontology to describe geometrical features. OntoCityGML uses the OpenGIS ontology to describe the LoD0Footprint (i.e., Level of detail 0 footprint) of a building and its wall surface area, while OntoFlood uses the OpenGIS ontology to describe the geographical area of the flood. By using standardised ontologies, these frameworks ensure compatibility between established geospatial representations and existing datasets, enabling interoperability between different types of information.

3. Methodology

3.1. Data sources

In this study, the routing functions, the isochrone generation functions, the critical path analysis, and the building matching function use OpenStreetMap (OSM) data, an open-source map collaboratively created by the community (OpenStreetMap contributors, 2017). OSM contains a wide range of geographic information about buildings, structures, and transportation network in the form of OSM elements (*i.e.*, points, lines and polygons). The features of these OSM elements are conveyed through OSM tags which describe the properties of these elements. OSM serves as the base map for these functions due to its open-sourced nature, easy usability, and integration with a wide range of open-source libraries such as pgRouting (pgRouting, 2023c) and osm2pgrouting (pgRouting, 2023b). Therefore this allows easy scalable solutions across different cities and countries.

To evaluate the population potentially at risk of flooding in King's Lynn, UK, the study utilises the OpenPopGrid, a population distribution dataset derived from the Office for National Statistics 2011 Census and OS OpenData (Murdock, Harfoot, Martin, Cockings, & Hill, 2015). To evaluate the population coverage of amenities in Pirmasens, Germany, Facebook Data for Good High-Resolution Population Density Maps is used (Tiecke et al., 2017). Additionally, the land use model is sourced from the municipal government in Pirmasens (Das Landesamt für Vermessung und Geobasisinformation (LVermGeo) Rheinland-Pfalz, 2023), it contains information about the designated purposes for the land, which can be for residential, industrial, sports, *etc*. The 3D CityGML data (Kolbe, Gröger, & Plümer, 2005) which describe the semantic representation of 3D city models and the 1-in-100 years flood raster data were obtained from the industrial partners.

3.2. Ontological representation

This paper introduces OntoIsochrone — an ontology to connect isochrone, road condition, transport mode, its originating source location, and its covered population data. The guiding principle in developing this ontology is to create a semantic representation of isochrones for geospatial analysis, as there has been no prior work done in this area.

The `iso:Isochrone` class is the key concept to represent isochrones and the `iso:originatesFrom` relationship connects it with an external geometry class `geo:Geometry` which its instance can be any elements with geographic information. The purpose of `iso:originatesFrom` predicate is to link each isochrone to its point of origin's ontological class. Additionally, since `iso:Isochrone` class contains geometry information, it is defined as a `rdfs:subClassOf` the `geo:Geometry` class, thereby enabling `iso:Isochrone` to be query-able via GeoSPARQL (Battle & Kolas, 2011). OntoIsochrone includes `iso:TransportMode` describes the transport mode used in calculating the area of reach within the isochrone, which can be Walking, Cycling, and Driving. The `iso:RoadCondition` describes the road condition used in isochrone calculation. The `iso:hasTimeThreshold` predicate defines the maximum duration within which all points in the isochrone can be reached from a source point. Furthermore, OntoIsochrone includes predicates to describe the detail of populations by gender, age, such as `hasMenPopulation`, `hasWomenPopulation`, `hasWomenofReproductiveAge`, `hasChildrenPopulation`, `hasYouthPopulation`, `hasElderlyPopulation`. These population predicates describe the underlying population data provided by Facebook's Data For Good (Tiecke et al., 2017).

Other relationships to isochrone are semantically defined within OntoIsochrone in Fig. 1. The implementation of the semantic representation of isochrone meta properties (*i.e.,* population, area of coverage, *etc*.) facilitates geospatial querying via GeoSPARQL (Battle & Kolas, 2011). GeoSPARQL enables the retrieval of geospatial data on the semantic web, allowing the creation of applications that can reason both general and geospatial information to provide cross-domain insight.

3.3. Agent-based implementation

The work in this paper utilises one input agent (*i.e.,* OSM Agent) and four output agents (*i.e.,* Routing Agent, Isochrone Agent, Travelling Salesman Agent, Network Analysis Agent). Two of these agents (*i.e.,* OSM Agent, Isochrone Agent) instantiate information and results back into the KG. These agents interact with the KG as depicted in Fig. 2. The OSM Agent retrieves data from the KG per OntoCityGML, derives building usage information from OpenStreetMap, and integrates it into the OntoBuiltEnv ontology. Routing Agent queries for areas of flood from OntoFlood and performs routing avoiding flooded area. Travelling Salesman Agent queries for the location of points of interest via OpenGIS ontology, evaluates whether these points of interest are flooded, and finds the fastest route to visit these points of interest once. Isochrone Agent queries for the location of points of interest via OpenGIS ontology, generates isochrones from that location based on the road conditions and transport mode, and subsequently instantiates the isochrone metadata back into the KG. Network Analysis Agent queries for the location of points of interest (*i.e.,* hospitals) via OpenGIS ontology and performs road network sensitivity analysis before and during flood to identify the important paths to reach the points of interest. To execute the algorithm of each agent, users initiate these agents through an HTTP POST request which can contain parametric inputs that can be agent configuration or locations.

3.3.1. OpenStreetMap Agent (OSM Agent)

The OSM Agent is designed to incorporate OSM data to augment instantiated 3D CityGML buildings. First, the agent maps the retrieved OSM tags to the OntoBuiltEnv concept which includes healthcare services (*i.e.,* clinics, hospitals, pharmacies), emergency services (*i.e.,* police and fire stations), recreational facilities (*i.e.,* drinking and eating establishments, hotels, sports facilities), residential buildings, industrial buildings, *etc*. This step processes the OSM tags into ontological concepts for improved knowledge re-usability.

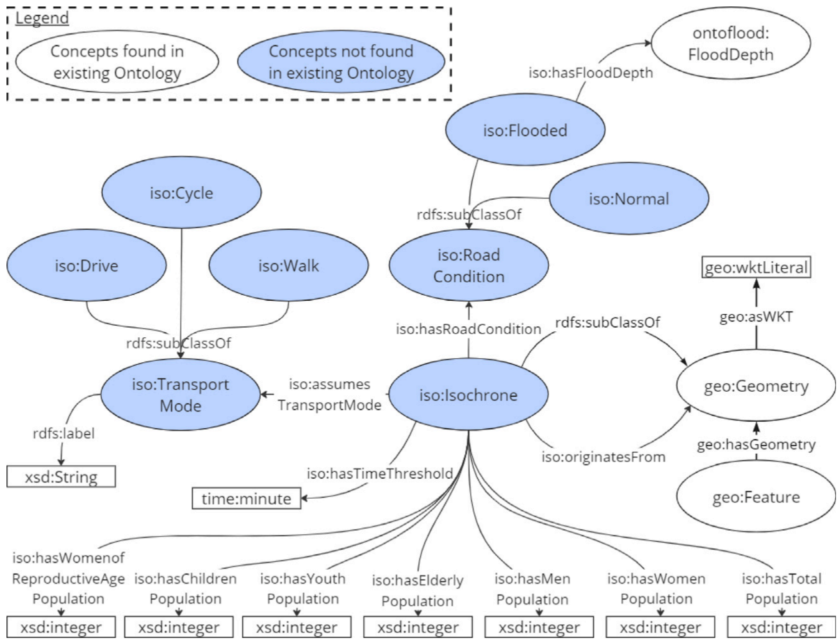


Fig. 1. OntoIsochrone describes the relationships associated with isochrone, this includes transport mode, road condition, demographics population, time threshold, and its connection with OpenGIS ontology. All referenced namespaces are declared in Appendix A.

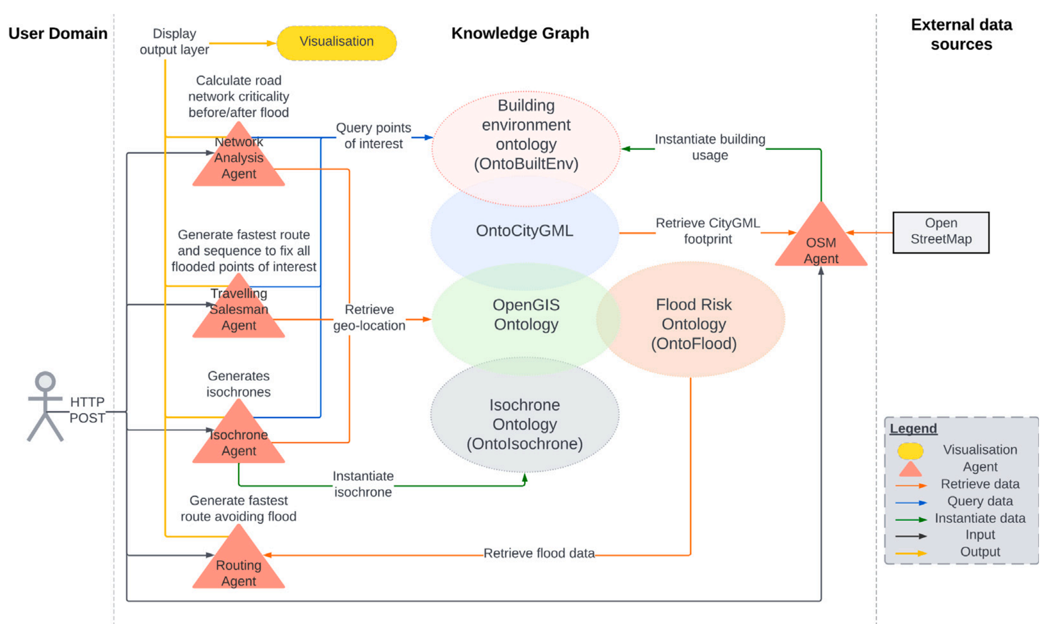


Fig. 2. Agents interplay on how OSM Agent, Routing Agent, Isochrone Agent, Travelling Salesman Agent, Network Analysis Agent dynamically derives and instantiates information from and into the knowledge graph.

Subsequently, the agent proceeds to geometrically identify and match the OSM data with 3D CityGML data by comparing the geometry of the OSM entries to the footprints of the 3D buildings. This procedure links and establishes the semantic connection between identical entities that are present on both datasets (*i.e.,* OSM data, 3D CityGML) which were previously unconnected. Following a successful matching between OSM data and 3D CityGML data, the agent computes the building usage share for each property usage type, as a single building can have multiple property usages. Each 3D CityGML building now has been augmented with the OSM data to provide richer insights into what the buildings are used for. For the untagged buildings to any OSM elements, the OSM Agent assigns building usages based on its corresponding municipal land use plots (*e.g.,* residential, industrial, religious facility, *etc.*) sourced from the local authority. A detailed UML diagram is provided in Fig. C.11 in the Appendix. To enhance the efficiency of mapping OSM data entries and municipal land use plot entries to OntoBuiltEnv, mapping tables were used to expedite the instantiation process from raw data to ontological terms, these tables can be found through the repository links located in Appendix D.

As OpenStreetMap data often comes with a flexible data model (unclear tags, free-text fields, etc.), the augmentation of 3D CityGML buildings with precisely defined classes of property usage of OpenStreetMap data through OntoBuiltEnv can overcome the challenges in the identification of certain points of interests. This facilitates the execution of the Isochrone Agent which needs the accurate geo-location of amenities to run accessibility analyses. Additionally, the augmented 3D

CityGML buildings with property usages can be reused for other future applications such as to perform electrical demand predictions using the City Energy Analyst based on building usage described in OntoBuiltEnv and its geometrical properties described in OntoCityGML (Fonseca, Nguyen, Schlueter, & Marechal, 2016).

3.3.2. Routing Agent

The Routing Agent provides context-aware routing solutions, meaning the agent derives optimised routing suggestions based on dynamic flood data instantiated via OntoFlood. These flood data can be provided by climate simulations, forecasting, or actual flood warnings. By overlaying the flood depth raster and the road network data, road segments that are affected by flood waters can be accurately identified, and excluded from emergency routing decisions.

Key functionalities of the Routing Agent are enabled by the open-source dynamic cost-based routing library pgRouting (Choosumrong, Raghavan, & Realini, 2010). pgRouting is a well-established tool for solving graph problems and is commonly used in emergency routing and decision planning applications (Choosumrong et al., 2019; Singh et al., 2015). Additionally, pgRouting has been utilised for locating the nearest shelter during flood emergency events (Wang, 2013), which emphasises its suitability for our research.

In pgRouting, road networks are divided into a graph network $G = (\{V\}, \{E\})$, consisting of a set of nodes (*i.e.,* V) and a set of edges (*i.e.,* E). The cost function on the edges can be modified programmatically and, subsequently, designates certain routes as inaccessible due to flooding. This is achieved by assigning a negative value associated with flooded roads. The Dijkstra's shortest path algorithm is then employed to find the shortest path to reach the destination while avoiding those flooded areas.

Routing Agent retrieves the flood depth raster via OntoFlood and identifies which roads are unusable for any given maximum inundation depth. Subsequently, the Routing Agent accepts user inputs with a start and end point to compute the fastest route for the maximum allowable wading depth. The resulting route is then displayed via the visualisation interface. Therefore, the augmentation of flood data from OntoFlood enables the identification of flooded routes, allowing Routing Agent to perform intelligent route planning while avoiding flood-affected areas. Routing Agent currently ingests 1-in-100-year flood projections for disaster planning purposes, real-time or historical flood raster data can be easily adopted as the agent ingests raster data.

3.3.3. Isochrone Agent

The Isochrone Agent functions by querying the location of the desired points of interest (POIs) from the KG and then generating areas of reach known as isochrone from these locations. The advantage of employing ontologies and KG for querying POIs lies in the ability to leverage semantically defined RDF data types. Using ontology concepts and defined classes (*i.e.,* Hospitals, Clinics, Police Stations, *etc*.) ensures precise POI locations, expedites search processes, achieves model re-usability, and eliminates data ambiguity effectively.

The Isochrone Agent algorithm is initiated by receiving inputs from the user specifying the desired time interval and segmentation length. The Isochrone Agent segments the road network based on the specified segment length and breaks down the road network into Delaunay triangulation polygon segments (Praliaskouski, 2018). Subsequently, the Isochrone Agent retrieves the location of POIs from the KG and calculates the travel time to each of these segments using pgRouting's *pgr_drivingDistance* function. Each triangle segments are assigned a calculated travel time from the POIs. Isochrones are formed by selecting the triangles segments which are less than the maximum time threshold specified. Subsequently, the Isochrone Agent calculates and maps the isochrone areas onto the population distribution data, thereby calculating the sum of the population covered. The resulting isochrones (*i.e.,* represented in the form of a multi-polygon) are displayed as an output on the visualisation interface. The isochrone information (*i.e.,* geometry shape, population covered, *etc*.) is then instantiated into the KG via Ontology-Based Data Access (OBDA) mapping (Bereta, Xiao, & Koubarakis, 2019), thereby exposing these geospatial data stored in a relational database to the KG, enabling cross-domain analysis with knowledge graph via GeoSPARQL. A detailed UML diagram is provided in Fig. C.12 in the Appendix.

Isochrone Agent allows the abstraction of using different modes of transport and road conditions (*e.g.,* Flooded or Normal) by specifying the characteristic of the road parameters in the EdgeTableSQL. EdgeTableSQL in pgRouting (pgRouting, 2023a) is a SQL statement that describes the road cost table used in pgRouting. It defines the allowable road network (*i.e.,* selecting roads that are not flooded). At the same time, this approach provides the flexibility to define any vehicle characteristics modelled through time costs. For example, the time cost for walking is represented by the road length divided by the average walking speed, similarly, the time cost for driving is represented by the road length divided by the speed limit. In this study, the driving speed is based on the maximum speed limit of the road, while the cycling speed is approximated to be 5.5 m/s and the walking speed is approximated to be 1.3 m/s. The approximation speed value for cycling from the average value of the available literature review on walking and cycling speed (Murtagh, Mair, Aguiar, Tudor-Locke, & Murphy, 2021; Pucher & Buehler, 2012). While real-world transport speed may vary due to factors like elevation, number of turns, traffic, *etc*., using an approximated value for the transport mode speed simplifies our analysis in this study.

3.3.4. Travelling Salesman Agent

The Travelling Salesman Problem is a computational challenge focused on determining the fastest route to visit all the nodes once in a graph network and return to the starting node (Noon & Bean, 1993). This problem is a commonly studied challenge in the field of operation research in its application for humanitarian relief logistics in disaster operations management (Crişan, Pintea, & Palade, 2017; Mosayebi, Sodhi, & Wettergren, 2021; Xu et al., 2021). Specialised vehicles such as amphibious vehicle, fire trucks, or helicopters are limited resources during a disaster, therefore a single vehicle may have to visit multiple locations in scenarios such as disaster rescue or infrastructure restoration. These locations/POIs, represented as nodes in a road network, should ideally be visited once to minimise delays.

The Travelling Salesman Agent identifies the flooded roads, retrieves the locations of POIs (*i.e.,* rescue locations or flooded infrastructure location) from the KG, and calculates the fastest route to reach these points while incorporating flood depth into the routing calculation. First, Travelling Salesman Agent queries and extracts crucial location data from the KG. Using the user input location as the starting point, the algorithm dynamically computes the optimal route to reach all points of interest and return to the original position while avoiding flood. The Travelling Salesman Agent evaluates the feasibility of reaching these locations via various modes of transport by using pgRouting's *pgr_tsp* function. If deemed unattainable, the algorithm appropriately flags and displays these unreachable POIs, and subsequently omits them from optimal route calculation. The fastest route and the sequence to visit all the points are then displayed as an output via the visualisation interface. By leveraging geospatial information and points of interest data organised in defined classes using ontologies, the Travelling Salesman Agent enhances the execution of travelling salesman problem by using KG to identify and locate the points of interest accurately.

3.3.5. Network Analysis Agent

Roads have varying importance based on their specific position within the transportation network (Gangwal, Siders, Horney, Michael, & Dong, 2023). Researchers have extensively examined road criticality to help in the prioritisation of infrastructure restoration after disasters (Gangwal et al., 2023). One method used for this purpose is the

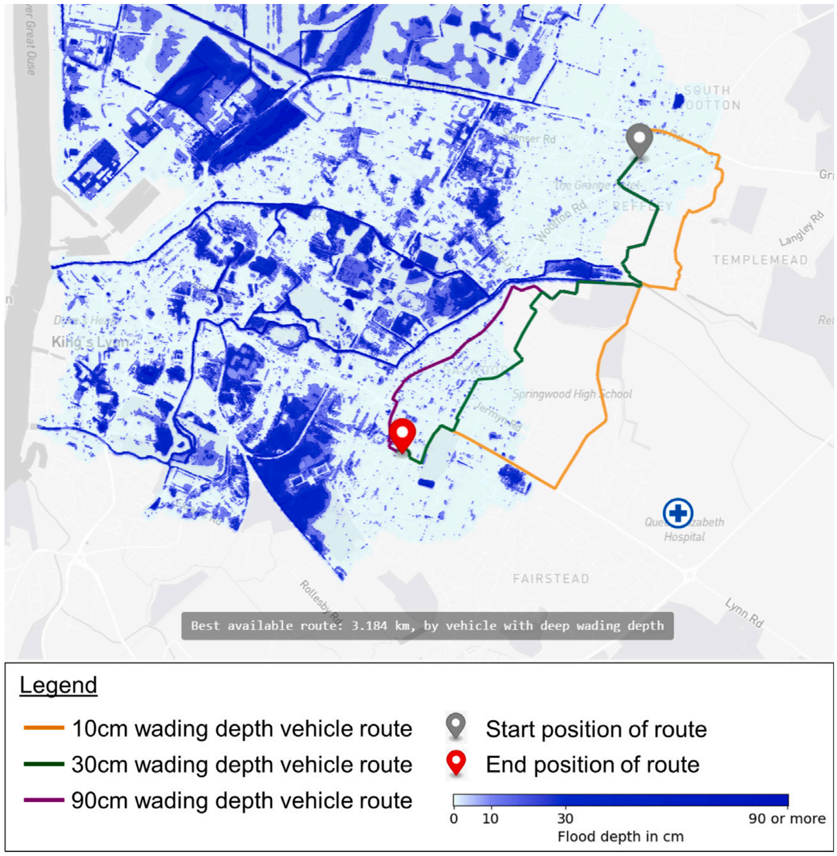


Fig. 3. Optimal route selection under flooded conditions where orange, green, and purple lines represent the fastest paths taken by vehicles with 10 cm, 30 cm, and 90 cm wading depth capabilities, respectively.

betweenness centrality measure which assesses the significance of roads in a network (Petricola, Reinmuth, Lautenbach, Hatfield, & Zipf, 2022; Tachaudomdach, Upayokin, Kronprasert, & Arunotayanun, 2021), this metric is often regarded as one of the most important metrics for prioritising recovery efforts in transportation infrastructures (Bhatia, Kumar, Kodra, & Ganguly, 2015). Betweenness centrality identifies the importance of a road based on how frequently it appears in the possible shortest paths from a particular node. Roads that are featured more frequently in these shortest paths are considered more important.

The Network Analysis Agent assesses the accessibility of critical amenities (*e.g.,* hospitals) by measuring the betweenness centrality metric before and during a flood. The Network Analysis Agent first retrieves the critical amenities location from the KG, and then computes the betweenness centrality metric for the road network from the critical amenities location. The changes in percentage in the betweenness centrality metric before and during a flood are displayed as an output via the visualisation interface. Through this direct comparison, critical pathways can be pinpointed by identifying the roads with the most significant percentage decrease in betweenness centrality before and during a flood. The implementation of KG and Network Analysis Agent facilitates seamless network criticality analysis for specific critical building types (e.g., hospitals, fire stations, and police stations). The Network Analysis Agent performs network criticality analysis using geospatial information and points of interest data categorised using ontologies, the agent uses knowledge graphs to accurately find and pinpoint the location of these important points of interest.

3.4. Computational resources

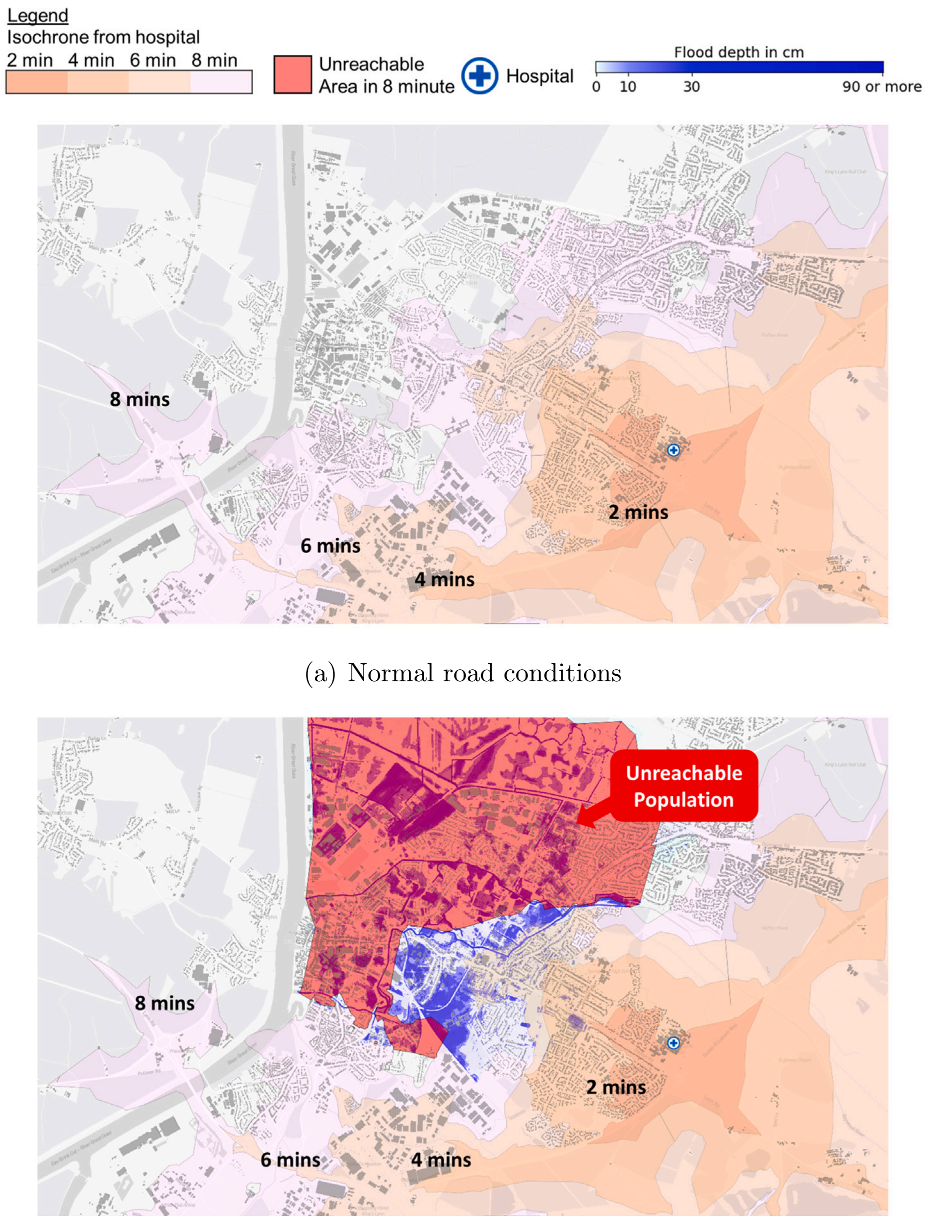
In this project, Blazegraph is employed as the triple store for the KG triples. To enhance its capabilities, Blazegraph is further complemented with virtual triples (OBDA) through Ontop (Calvanese et al., 2017) which utilise PostgreSQL as the underlying database. All these services are deployed in a containerised fashion to support platform-agnostic deployment. To support data assimilation and intra-stack communication, tools like GDAL and NGINX are included. Lastly, for visualisation purposes, Geoserver and a custom TWA visualisation interface are provided.

In King's Lynn, the urban resilience tech stack runs on a local server featuring an Intel Xeon CPU @ 2.00 GHz and 128GB RAM. Meanwhile, in Pirmasens, the 15-minute city tech stack operates on a Digital Ocean server with 32 GB RAM, and 100GB HDD. The number of triples for the King's Lynn tech stack is around 11 million. While the number of triples for the Pirmasens tech stack is around 8.7 million. The types of data incorporated in the tech stacks include raster datasets such as elevation, flood, and population data, alongside vector geometries such as isochrone polygons, CityGML buildings, landplots, and road networks. These deployments cater to varying geographical extents, with King's Lynn covering approximately 15.5km^2 and Pirmasens spanning around 61.7km^2.

4. Use cases

To demonstrate TWA's potential as a versatile flood disaster response tool, King's Lynn a mid-size coastal town in the East of England, has been selected for a proof of concept for the Urban Resilience use case (refer to Section 4.1). The selection of King's Lynn is strategic, considering the town's vulnerability to coastal flooding and the wealth of available open data in the UK.

Additionally, to illustrate TWA's capabilities as a planning tool for 15-minute city planning, a small town of 40,000 people — Pirmasens in Rhineland-Palatinate, Germany, has been chosen (refer to Section 4.2).



(a) Normal road conditions

(b) Flooded conditions with 30 cm wading depth vehicle

Fig. 4. Demonstration of the capability of TWA to create isochrone maps for access times from hospitals. (a) represents the isochrone maps in 2-minute intervals from Queen's Elizabeth Hospital. (b) represents the isochrone maps under flooded conditions and outlines the area of unreachable population in 8 min.

Facing a dwindling population, the local municipal government in Pirmasens is actively seeking improved methods for accurate resource allocation. The overarching goal is to ensure that the town's amenities can adequately cater to all residents. This requires streamlining and integrating fragmented data sources to enable the municipality to make well-informed decisions for optimised resource distribution.

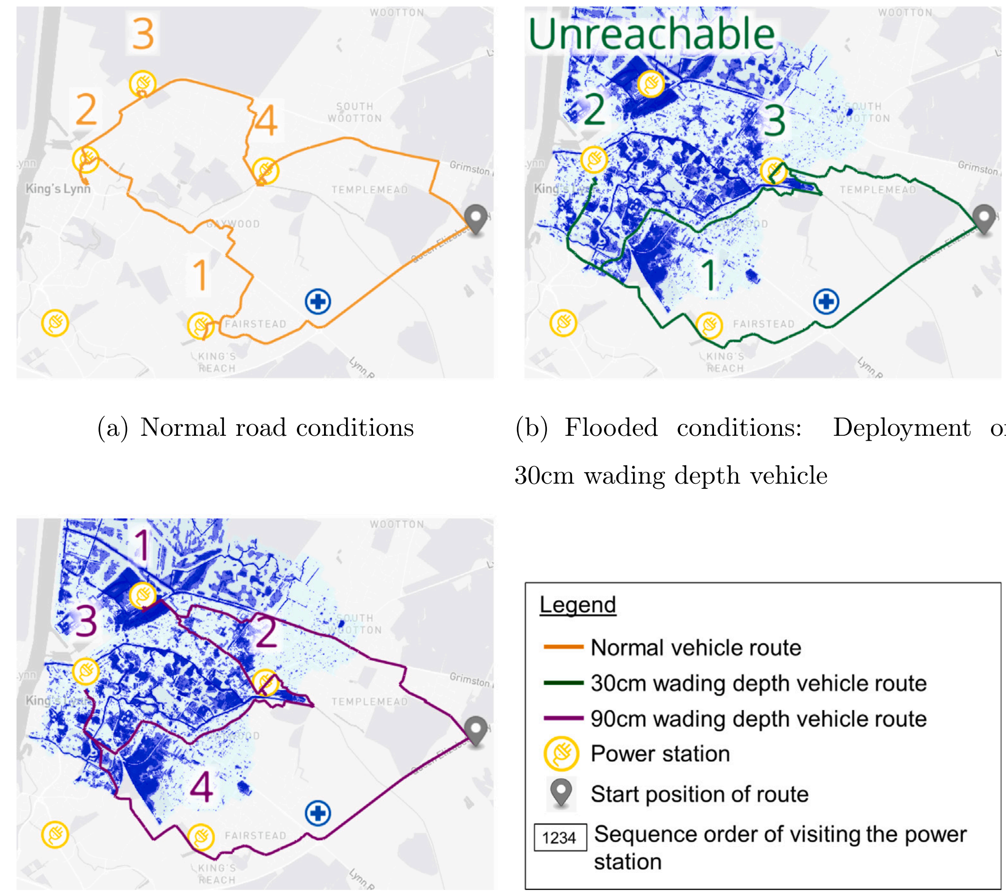
4.1. Urban resilience

In King's Lynn, 8% of the town's population is aged 75 years or above, and approximately 7.7% of the total town's population is physically challenged in their day-to-day life (Office for National Statistics, United Kingdom, 2021). This highlights the need to ensure rapid, efficient, and specialised assistance for these vulnerable demographics during severe weather and disaster events. During such situations, rapid deployment of healthcare and rescue services is required. Loss of local medical services and amenities combined with restricted accessibility to the people at risk are factors that will drastically increase the mortality rate (Akbari, Farshad, & Asadi-Lari, 2004). Therefore, advanced preparedness and anticipatory actions are required to provide a quicker and more effective response during a sudden disaster.

4.1.1. Context-aware routing

In the event of floods, specialised vehicles such as helicopters, boats, ambulances, fire trucks, and high-water trucks are limited resources. These vehicles each have distinct operational speeds, allowable water wading depth, and deployment times. It is essential to effectively coordinate and allocate resources according to various vehicles and the locations requiring rescue.

One way to achieve this is by integrating flood depth level dynamically into routing calculations. Fig. 3 illustrates the effectiveness of this approach, showcasing the selection of a route that circumvents flooded



(a) Normal road conditions

(b) Flooded conditions: Deployment of 30cm wading depth vehicle

(c) Flooded conditions: Deployment of 90cm wading depth vehicle

Fig. 5. Fastest route to restore power stations during normal road conditions and flooded conditions.

areas, leading to a swift and efficient emergency response by a fire truck.

4.1.2. Accessibility analysis (healthcare)

The key to reducing the consequences of disasters is being able to respond and prepare for them effectively (Alexander, 2015). This includes evaluating the loss of physical access to critical infrastructure and identifying the population that will likely be impacted. The Isochrone Agent addresses this by identifying the area of reach from any points of interest under flooded conditions. Isochrones help to identify who is covered by certain critical infrastructure and what share of the population is potentially unreachable.

The agent retrieves the locations of critical infrastructure such as hospitals, police stations, fire stations, *etc*. Subsequently, the agent generates polygons with the reachable area from each location of interest within a certain time threshold, so-called isochrone maps.

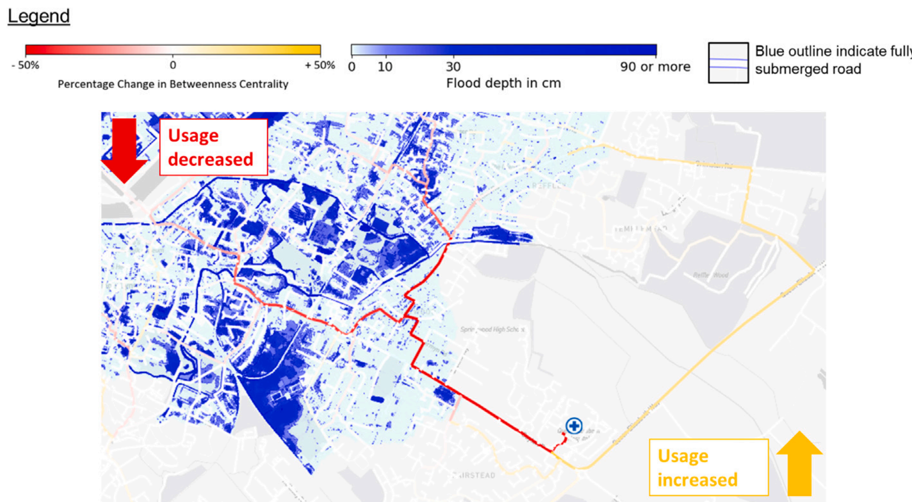
For example, an isochrone map that depicts the service area under normal conditions from the hospital in King's Lynn is shown in Fig. 4(a). During severe weather, certain areas might become inaccessible due to flood. Insights on the segment and the share of the population which is likely unreachable within a 10-minute timeframe can be derived through the consideration of the population density as seen in Fig. 4(b). By gaining an understanding of the operational region of essential infrastructures, it becomes possible to pinpoint inaccessible areas. This knowledge can then be used to determine strategic measures needed in emergency service planning to guarantee comprehensive coverage, avoiding blind spots. Currently, the flood scenario in Fig. 4(b) uses a 1-in-100-year flood scenario. This approach could be applied to multiple projected flood scenarios.

4.1.3. Travelling Salesman problem

Flooded roads block off available paths and increase travel time to critical infrastructure (*i.e.,* power stations). Failures of multiple infrastructures caused by disaster events often cannot be restored at the same time because of the limitation of the recovery resources, such as recovery vehicles (Ma et al., 2015). The objective of reducing the travel time to restore these critical infrastructures in case of failure can be modelled as a Travelling Salesman Problem.

The application of the Travelling Salesman Agent supports the fastest restoration of failed power stations submerged due to flooding by minimising the travel time. UK Power Network's power stations (UK Power Networks, 2023) are first identified if flooded. Travelling Salesman Agent then calculates the fastest route to restore these flooded power stations while incorporating flood depth and vehicle water depth wading capability, as seen in Fig. 5. The optimised route is determined and the sequence for visiting these power stations is outlined. Each station is visited once only to reduce delay before returning to the starting node which typically represents a resource depot (Wang, Sarker, Mann, & Triantaphyllou, 2004).

This strategy ensures a reactive response that minimises recovery time during a disaster. On top of that, additional information such as the hierarchical importance of specific infrastructures and the consideration of cascading network effects can be integrated into the travelling salesman analysis for a more holistic approach. Similarly, this approach



(a) Changes in betweenness centrality before and during flood

(b) Flooded unusable roads (*i.e.*, outlined by blue) with high reduction in betweenness centrality metric (*i.e.*, coloured with brighter red) indicates the more critical roads to be prioritised

Fig. 6. The percentage change in betweenness centrality is visually represented using red and orange hues. The red lines denote paths that experience a reduction in the betweenness centrality metric, signifying a diminished role in the total shortest paths before a flood. The orange lines denote paths experiencing an increase in the betweenness centrality metric, highlighting paths that have become more prominent during a flood in the overall of the total shortest paths originating from the critical POI.

can also be applied for relief goods distribution or mobile medical care (Sheth et al., 2019).

4.1.4. Critical path analysis

In post-disaster situations with a large portion of damaged transportation networks, such as roads and bridges, the task of coordinating recovery resources systematically and building a reconstruction plan robustly can be challenging (Ghannad, Lee, & Choi, 2021). Administrators often need to organise limited resources in planning their subsequent and rapid reconstruction as poor transportation networks could severely limit the recovery efforts (Freckleton, Heaslip, Louisell, & Collura, 2012). Critical path analysis can provide a guideline on how to prioritise reconstruction efforts after a disaster. It can determine which roads or pathways should be repaired first, ensuring a smooth, continuous, and efficient overall disaster response. Furthermore, this ability allows us to identify which road should be prioritised to be kept clear from traffic or, if possible, even flood waters to streamline disaster response.

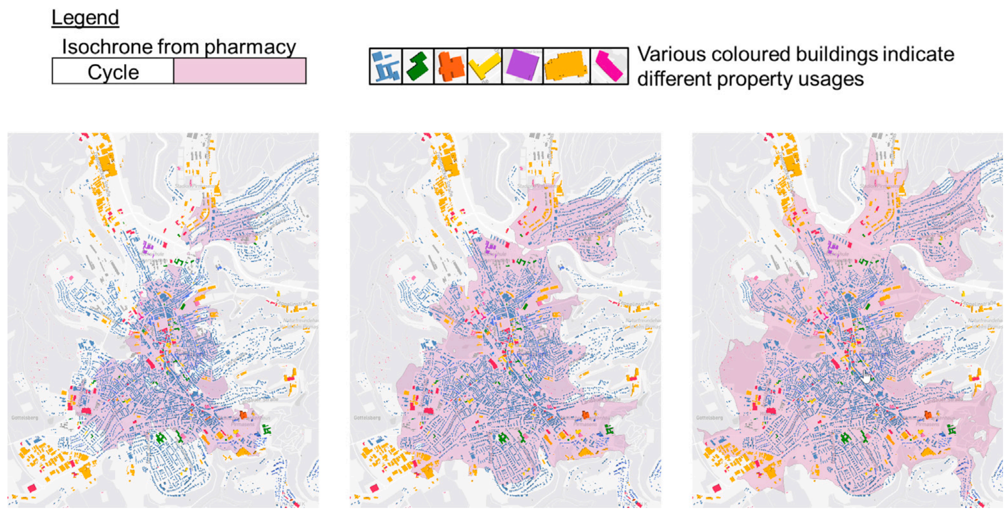
Using this approach, the Network Analysis Agent queries and retrieves accurate geo-locations of critical infrastructure (*i.e.,* hospitals, police stations) from the KG. Subsequently, the agent conducts a sensitivity analysis both before and during a flood, as depicted in Fig. 6(a). Infering from this analysis, the paths that experience a huge decrease in the betweenness centrality metric (*i.e.,* represented in red) should be prioritised for restoration. On the other hand, the paths that experience an increase in the betweenness centrality metric (*i.e.,* represented in orange) become more relevant and these paths should be kept clear from flood or traffic during flood events. In Fig. 6(b) paths that are rendered unusable by the flood were outlined, and the most important paths can be identified and prioritised for restoration.

Overall, the proposed Network Analysis Agent can effectively guide rescue and repair efforts during and after a disaster to ensure continuous accessibility to essential services. As a result, improved city planning that integrates strong, robust transportation network systems and flood resilience can be attained.

4.2. Improving the livability of cities

4.2.1. Status quo assessment

TWA aims to provide status quo assessments that offer city officials a means to analyse the current amenities coverage of their cities.



(b) 5 minutes (c) 10 minutes (d) 15 minutes

Fig. 7. Isochrones via cycling from all pharmacies in Pirmasens.

Isochrone Agent can retrieve the accurate locations of buildings with a certain usage (category), allowing it to generate isochrones based on the gathered information. The critical points of interest used by the Isochrone Agent which are crucial for a 15-minute city include health-care services (*e.g.,* clinics, hospitals, pharmacies), emergency services (*e.g.,* police and fire stations), recreational facilities (*e.g.,* drinking and eating establishments, hotels, sports facilities), residential buildings, industrial buildings, transportation facilities, educational institutions, retail establishments, offices, and banks. Through this accessibility analysis, local municipal governments can evaluate the current coverage of city services for its residents, for example, the area of coverage by all the pharmacies in Pirmasens as shown in Fig. 7.

Similarly, isochrones can be generated for various transport modes such as walking, cycling, and driving from various amenities that include, but are not limited to, banks, schools, and hospitals, as shown in Fig. 8. By doing so, gaps and blind spots in the coverage area of these amenities can be effectively identified.

On top of that, accessibility can be assessed to evaluate if the entire population is equally covered. City officials can investigate whether certain essential service coverage correlates with certain socio-economic parameters such as wealth, age, or ethnicity. This assists governments in discovering whether a particular demographic group is lacking in accessibility to services which contributes towards social inequities within the urban fabric. It provides tools for the administrators to draw findings from the correlation analysis from parameters such as the housing prices as described through OntoBuiltEnv and accessibility as described through OntoIsochrone in the knowledge graph domain. Studies have found that the accessibility to local amenities such as dining, shopping, health services, *etc*., can have different impacts on the property prices (Beracha, Gilbert, Kjorstad, & Womack, 2018; Yuan, Wei, & Wu, 2020; Zhang, Xiong, & Wei, 2022).

These accessibility studies enable local municipal government to build the right infrastructure at the right location, reaching the intended demographics and filling previous coverage gaps. An interesting observation in the analysis is that contrary to conventional beliefs, residing in the city centre does not necessarily guarantee proximity to essential amenities as shown in Fig. 9, in which the accessibility to pharmacies is used as an example. The coverage of 15 min walking isochrone from all pharmacies in Pirmasens does not overlap at the city centre. This challenges the common assumption that central locations inherently provide convenient walking access to most amenities. Hence, such observations underscore the need for a comprehensive urban planning tool that allows current status quo monitoring.

4.2.2. Scenario planning

TWA can address hypothetical scenarios that can be applied to strategic infrastructure development. Some of the questions TWA can answer include: (1) Where to place the next amenities? (2) Which land plot can it be built on? (3) What will be the new population coverage?

By strategically placing and validating the placements with underlying zoning regulations, TWA can assist governments in effective service planning. Some important aspects include planning emergency services through determining the best locations for critical infrastructures (*i.e.,* fire stations, police stations, emergency medical services) to minimise response times. Additionally, TWA can also analyses residents access to green spaces and recreational areas to identify areas in need of additional green infrastructure.

Overlaying information on the accessibility analysis with population density oftentimes can help spatial planners in designing solution-oriented upgrading programmes to pinpoint residential areas with poor access (Gutting, Gerhold, & Rößler, 2021). As seen in Fig. 10 which uses pharmacy as an example the placement of pharmacy was optimised based on the allowed zone and population distribution. Through the integration of land plots in TWA, TWA possesses the capability to improve coverage while adhering to the land plot regulations. Due to the limitations of open data, current population distribution estimates by demographics are based on census data (Tiecke et al., 2017). When more granular population distribution by socio-economic attributes is provided, better cross-domain insights can be achieved. This includes measuring the accessibility of medical healthcare services to reach the intended group such as the elderly population, children, or where to place the mobile vaccination centre.

4.3. Core capability of TWA

TWA exhibits versatility by supporting both dynamic and static planning, as demonstrated in the use cases outlined in Section 4.1 and Section 4.2. It supports dynamic planning through context-aware routing: TWA interacts with users to acquire location information, enabling the specification of origin and destinations for routing while dynamically considering factors such as flood avoidance. Two key agents, Routing Agent and Travelling Salesman Agent, intelligently recompute optimal routes based on the user input locations. These functionalities can enhance the agility and responsiveness of emergency response teams in disaster situations.

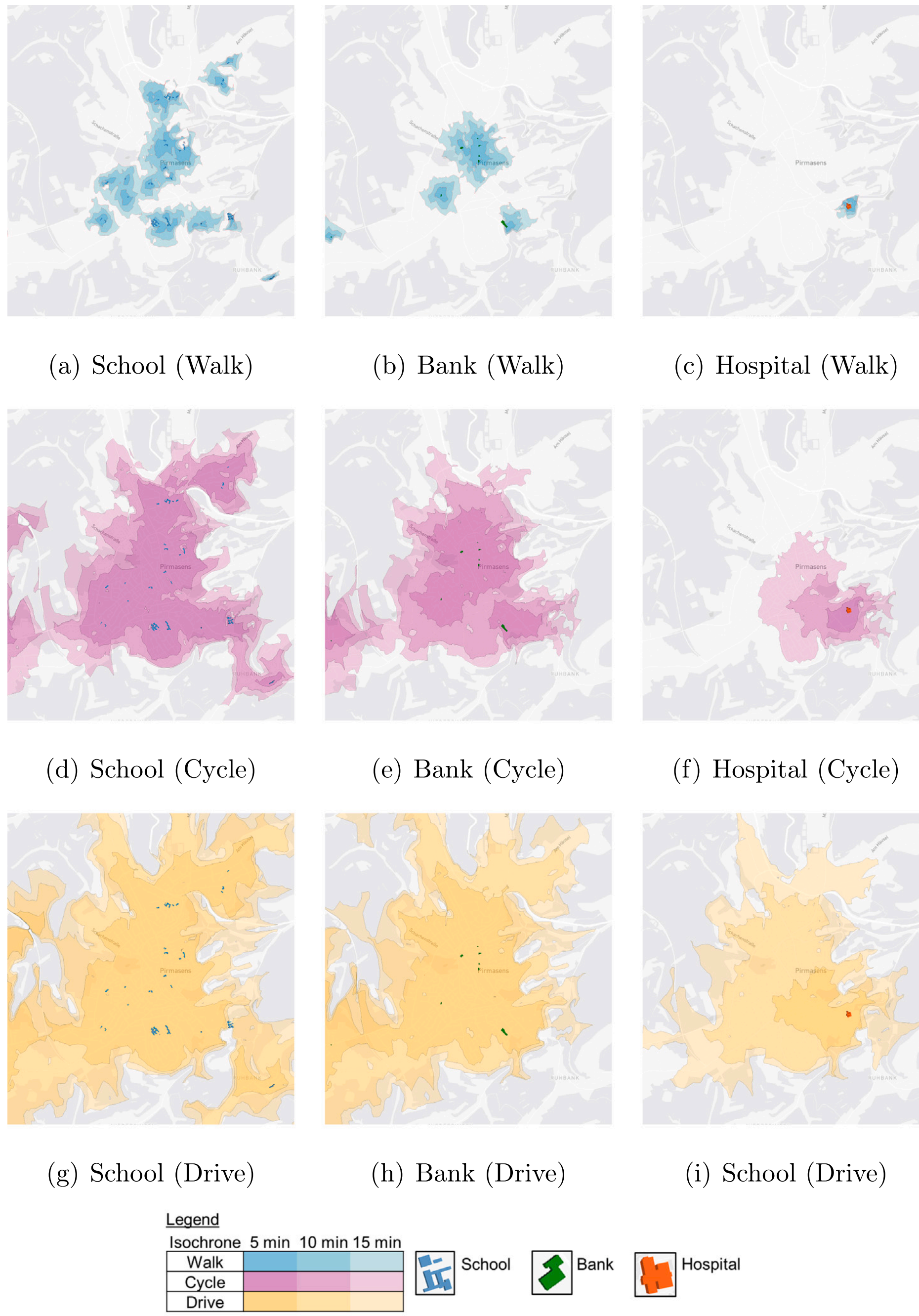


Fig. 8. Isochrones illustrating 5, 10, and 15 min travel time boundaries from the schools, banks, and hospitals, each colour-coded to represent specific modes of transportation: blue for walking, purple for cycling, and yellow for driving.

The static planning aspect of TWA is characterised by its ability to perform accessibility analysis and critical path analysis. In terms of emergency response, Isochrone Agent can effectively identify the area of reach from critical infrastructures and at the same time, identify blind spots and the area out of reach. While the Network Analysis Agent can identify important paths in a road network before and during a disaster. TWA can be applied in (1) correlation analyses to evaluate if household income correlates with factors such as accessibility to hospitals, schools (2) demographic insights through gaining a comprehensive understanding of population distribution by considering parameters like age, race, and income in specific geographic areas. A few examples of such analyses are as below, whereby the SPARQL queries can be found in Appendix B:

1. What is the average estimated building price within a 2-minute driving radius of the hospital in King's Lynn?
   Answer: Average £308,234 (based on 299 buildings in the area)
   Execution Time: 662.84 s
2. Find all buildings with a gross floor area between 80-120 m^2 within 2 min driving from the hospital in King's Lynn?
   Answer: 108 buildings (count shown instead of individual instances for readability)
   Execution Time: 266.28 s
3. What is the total number of elderly population per pharmacy within 10 min of walking?
   Answer: 106 elderly people per pharmacy
   Execution Time: 0.10 s
4. What is the name of the 3 tallest schools in Pirmasens?

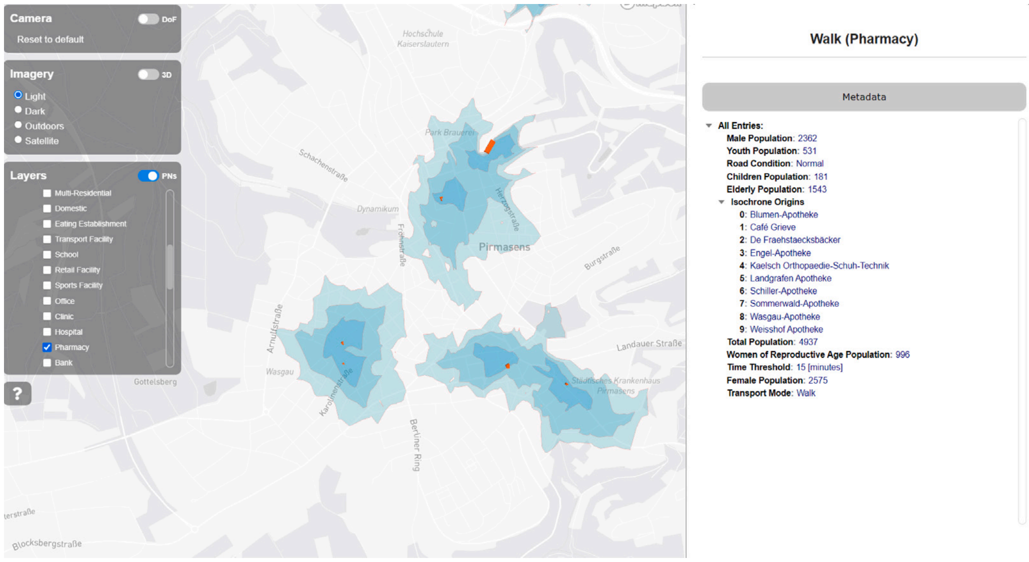
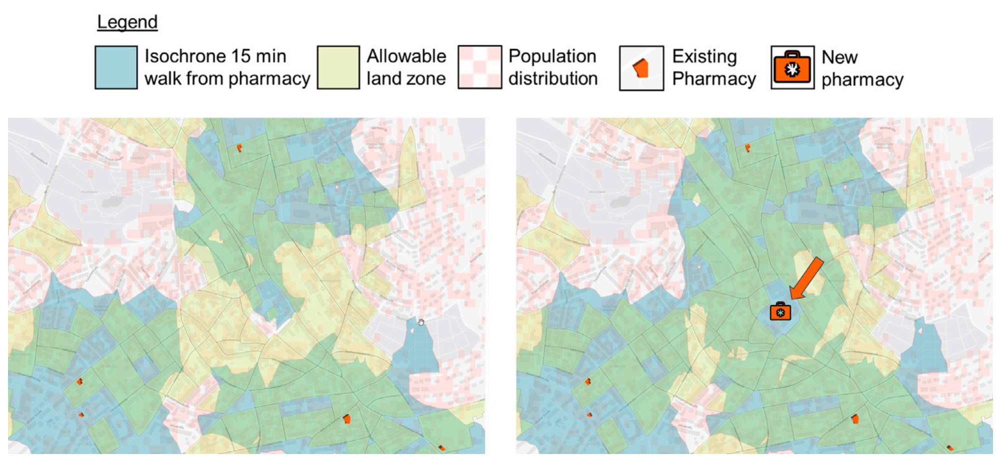


Fig. 9. User Interface of TWA Visualisation Framework which details the metadata of isochrones (*i.e.,* population covered by demographics, the building names of isochrone origination source, *etc.*)



(a) 15 minutes walk isochrone from pharmacies in Pirmasens

(b) Improvement in pharmacy accessibility via walking through the addition of a pharmacy at the allowable land plot zone in city centre

Fig. 10. Infrastructure planning through the multi-factor considerations where red grids represent population distribution, yellow polygons represent allowable zones for pharmacies to be built, blue represents 15-minute walk isochrone from pharmacies.

Answer: (1) Grundschule Wittelsbach - 27.03 m, (2) Landgraf Ludwig Realschule plus - 24.53 m, (3) Matzenbergschule - 24.34 m
Execution Time: 0.854 s

5. What is the area not covered by pharmacies within 15 min of walking in Pirmasens?
Answer: Provided in Fig. 9

Through TWA, these competency questions can answer some of the pressing policy planning challenges, for example, walkable cities for children that necessitate designing urban environments that are inclusive and conducive to walking, particularly for children residing in proximity to schools. Another applicable scenario includes environmental agencies whose vision is to achieve accessibility for all residential houses to be located within 10 min walking from parks (NParks, 2020). Addressing these challenges in policy planning is a crucial stride in the direction of creating a 15-minute city.

While one might argue that Geographic Information Systems (GIS) can easily address individual use cases, however leveraging solely on GIS often leads to less optimal outcomes due to restricted data silos that only encode just one domain/perspective (Janowicz et al., 2022). The distinctive advantage of leveraging a knowledge graph lies in its capability for cross-domain analysis while at the same time transcending geo-location dependencies. By serving as a gateway to various data sources, the knowledge graph approach facilitates scalability across diverse data sets, effectively overcoming data ambiguity. This results in a robust system characterised by high re-usability and extendability.

5. Conclusions

Knowledge graph driven technology like The World Avatar has the ability to connect and integrate flood data, OpenStreetMap data, land plot data, and 3D CityGML buildings data, through ontologies. These ontologies create semantic representations to facilitate machine-readability and automation, enabling cross-domain analyses to support emergency response and urban planning. In this paper, OntoIsochrone is introduced to semantically describe isochrone and its connecting relationship with transport mode, road condition, time threshold, and its originating source. We have developed a set of connected agents, working in conjunction with knowledge graphs, to support intelligent flood routing to avoid flooded roads and hence decrease risk for emergency personnel, optimise routing for post-disaster rectifications, perform accessibility analyses under flood situations using isochrones, as well as critical path analyses to streamline recovery efforts using betweenness centrality metric. In King's Lynn, we have integrated multiple domains to achieve interoperability, this includes geospatial data (i.e., CityGML buildings with property prices), simulated models (i.e., 1-in-100-year flood projections), social demographics (i.e., population distribution), healthcare accessibility (i.e., isochrones from hospitals).

In urban planning, the implementations of agents and OntoIsochrone ontology as part of The World Avatar extend its capability to support 15-minute city planning. This involves using isochrone to measure the reach and population coverage of essential amenities by different timescales and different transport modes such as walking, cycling, and driving. This ability is further extended to scenario planning which enables the optimisation of amenities placements to improve areal coverage while complying with land use regulations. At the same time, the application of ontologies to represent building information, isochrones, and populations, facilitates cross-domain correlation analysis and provides valuable insights. This approach enables planning capability across areas that represent buildings (i.e., CityGML buildings and property usages derived through OpenStreetMap), urban planning regulations (i.e., land plots), social demographics (i.e., population distribution), urban accessibility (i.e., isochrones derived from various transport mode).

Future work includes simulating and incorporating traffic flow under flood conditions to enhance the accuracy of informed routing as flooding will often lead to congestion due to lesser accessible roads. Potential future applications include using The World Avatar to perform route avoidance for other disasters such as forest fires or plumes. There are ongoing research efforts investigating methods to implement a dynamic plume model in routing avoiding forest fire (Wang & Zlatanova, 2019; Wang, Zlatanova, Moreno, Van Oosterom, & Toro, 2014). With The World Avatar's fully developed plume dispersion modelling considering real-time weather data (Eibeck et al., 2019), The World Avatar could be a suitable ecosystem to integrate an end-to-end solution that combines hazard simulation results to support routing during disasters. Additionally, The World Avatar currently ingests flood projections as raster data, our approach can be incorporated with real-time data for immediate disaster response or retrospective data for historical analysis. Other than that, future research could incorporate dynamic population behaviour into analysis, and also investigate the application of knowledge graphs in analysing the impacts of disasters on urban resilience.

City planning projects involve significant investments, therefore there should be an effort to emphasise reusing and extending existing data structures. Knowledge graph connects data, reuses ontologies, and eliminates data ambiguity. The practical deployment of The World Avatar in both King's Lynn, United Kingdom, and Pirmasens, Germany, showcases its location independence as an open data model. Simultaneously, the implementation of knowledge graphs and an agent-based approach in tackling two different scenarios: urban resilience and 15-minute city demonstrate the versatility and interoperability advantages of this method for cross-domain applications.

CRediT authorship contribution statement

Shin Zert Phua: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Markus Hofmeister:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Yi-Kai Tsai:** Validation, Software, Investigation, Formal analysis, Data curation. **Oisín Peppard:** Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Kok Foong Lee:** Writing – review & editing, Visualization, Validation, Software, Data curation. **Seán Courtney:** Software. **Sebastian Mosbach:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Jethro Akroyd:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Markus Kraft:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All the codes developed are available on The World Avatar GitHub repository: https://github.com/cambridge-cares/TheWorldAvatar.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to enhance the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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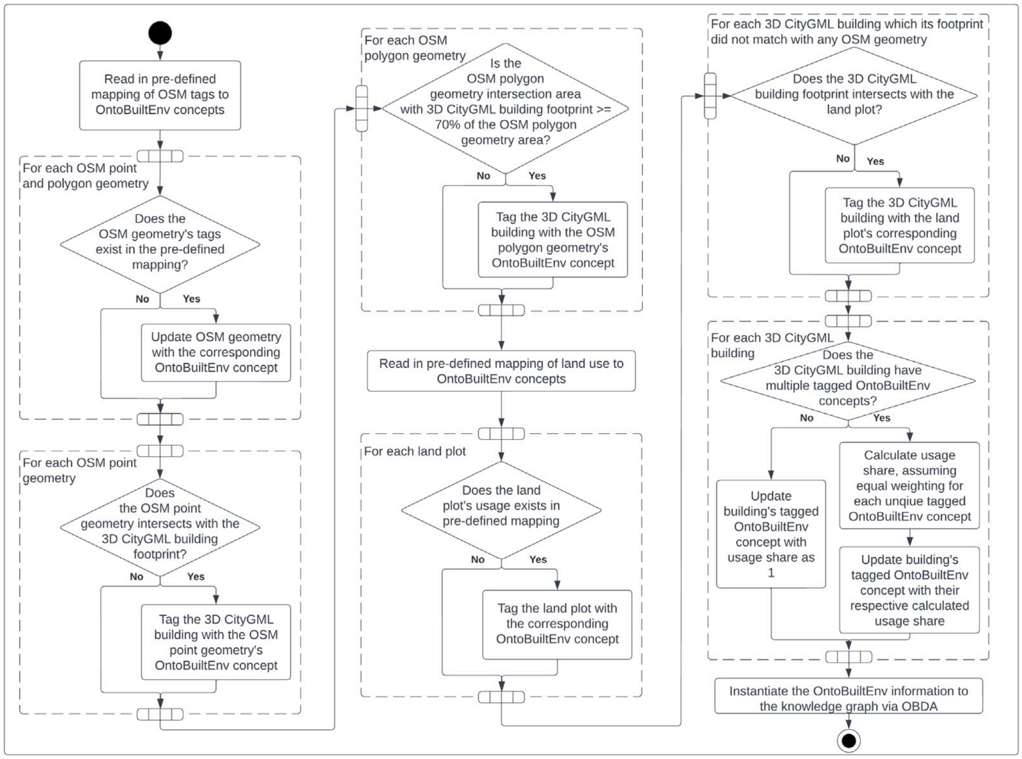

Fig. C.11. OSMAgent UML diagram.

Appendix A. Namespaces

rdfs: <http://www.w3.org/2000/01/rdf-schema#>
geo: <http://www.opengis.net/ont/geosparql#>
geof: <http://www.opengis.net/def/function/geosparql/>
om: <http://www.ontology-of-units-of-measure.org/resource/om-2/>
dabgeo: <http://www.purl.org/oema/infrastructure/>
obe: <https://www.theworldavatar.com/kg/ontobuiltenv/>
xsd: <https://www.w3.org/2001/XMLSchema#>
iso: <https://www.theworldavatar.com/kg/ontoisochrone/>
bldg: <https://schemas.opengis.net/citygml/building/3.0/>
ontoflood: <https://theworldavatar.com/kg/ontoflood/>
time: <http://www.w3.org/2006/time#>

Appendix B. Competency questions SPARQL queries

The following SPARQL queries can be used to answer the stated competency questions. It is important to note that Ontop is employed for the processing of GeoSPARQL queries. Hence, the placeholder `<ontop>` needs to be substituted with the actual Ontop endpoint before executing the queries.

Listing 1: What is the average estimated building price within a 2-minute driving radius of the hospital in King's Lynn?

```
SELECT (COUNT(?bldg) as ?buildings) (AVG(?price) as
     ?average)
WHERE {
?bldg obe:hasMarketValue ?mv .
?mv om:hasValue ?value .
?value om:hasUnit om:poundSterling ;
       om:hasNumericalValue ?price .
SERVICE <{ontop}> {
    ?bldg a dabgeo:Building ;
          geo:hasGeometry/geo:asWKT ?bldg_loc .
    ?iso a iso:Isochrone ;
        iso:hasRoadCondition ?rc ;
        iso:hasTimeThreshold "2"^^<http://www.w3.
            org/2006/time#minute> ;
        iso:originatesFrom ?poi ;
        geo:hasGeometry ?geometry .
    ?rc a iso:Normal .
    ?poi a obe:Hospital .
    ?geometry geo:asWKT ?polygon .
    FILTER ( geof:sfWithin(?bldg_loc, ?polygon) )
    }
}
```

Listing 2: Find all buildings with a gross floor area between 80-120 m^2 within 2 minutes driving from the hospital in King's Lynn?

```
SELECT ?bldg
WHERE {
?bldg rdf:type dabgeo:Building ;
      obe:hasTotalFloorArea ?area .
?area om:hasValue ?value .
?value om:hasUnit om:squareMetre ;
       om:hasNumericalValue ?num_val .
FILTER (?num_val >= 80 && ?num_val <= 120)
SERVICE <{ontop}> {
    ?bldg geo:hasGeometry/geo:asWKT ?bldg_loc .
    ?iso a iso:Isochrone ;
        iso:hasRoadCondition ?rc ;
        iso:hasTimeThreshold "2"^^<http://www.w3.
            org/2006/time#minute> ;
        iso:originatesFrom ?poi ;
        geo:hasGeometry ?geometry .
    ?rc a iso:Normal .
    ?poi a obe:Hospital .
    ?geometry geo:asWKT ?polygon .
    FILTER ( geof:sfWithin(?bldg_loc, ?polygon) )
    }
```

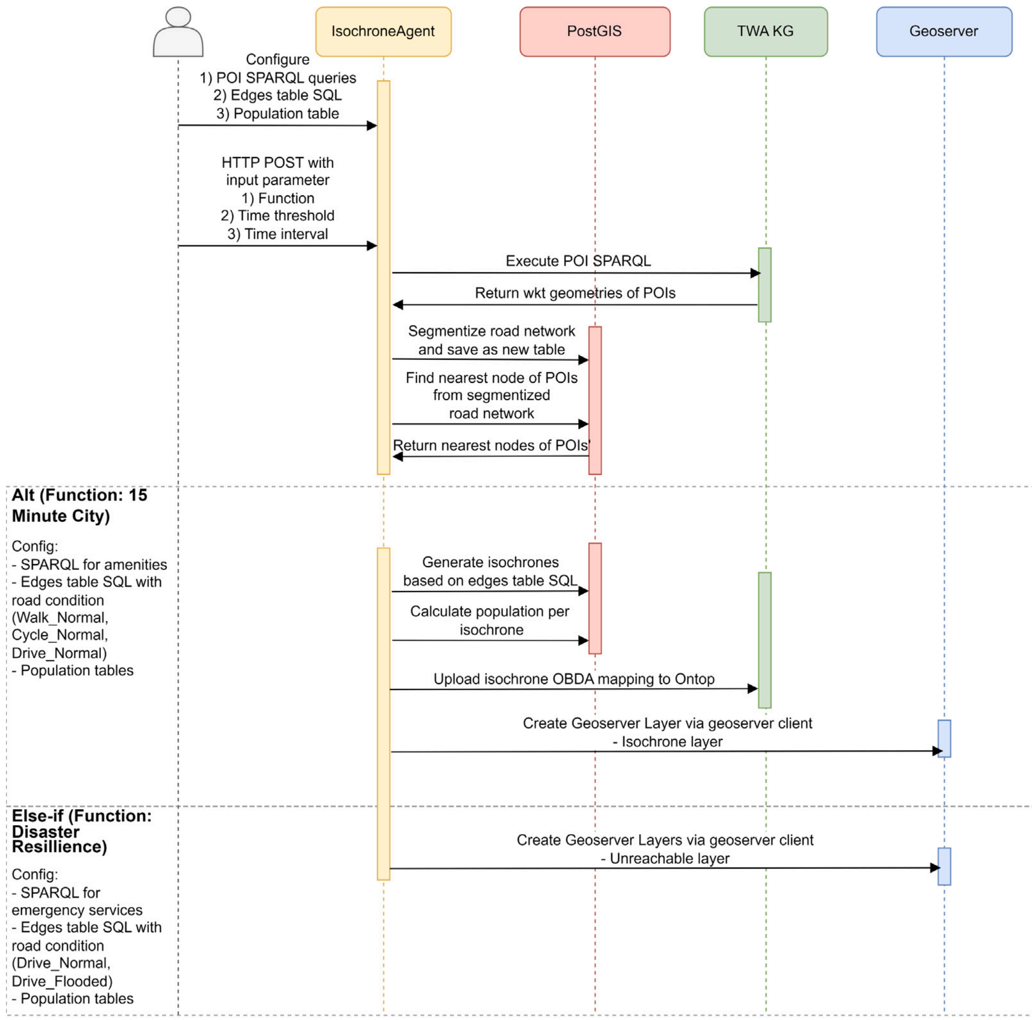

Fig. C.12. Isochrone Agent UML diagram.

```
}
```

Listing 3: What is the total number of elderly population per pharmacy within 10 minutes of walking?

```
SELECT ?elderlyPopulationPerPharmacy
WHERE {
  SERVICE <{ontop}> {
    SELECT (ROUND(?elderlyPopulation/?pharmacyCount
        ) as ?elderlyPopulationPerPharmacy) WHERE {
      {
        SELECT (COUNT(DISTINCT ?building) as ?
            pharmacyCount) WHERE {
          ?isochrone a iso:Isochrone .
          ?isochrone iso:originatesFrom ?building.
          ?building a obe:Pharmacy.
        }
      }
      {
        SELECT ?elderlyPopulation WHERE {
          ?isochrone a iso:Isochrone .
          ?isochrone iso:originatesFrom ?building.
          ?building a obe:Pharmacy.
          ?isochrone iso:hasTimeThreshold "10"^^<
              http://www.w3.org/2006/time#minute>.
          ?isochrone iso:assumesTransportMode ?
              transportMode.
          ?transportMode a iso:Walk.
          ?isochrone iso:hasElderlyPopulation ?
              elderlyPopulation.
        }
        GROUP BY ?elderlyPopulation
      }
    }
  }
}
```

Listing 4: What is the name of the 3 tallest schools in Pirmasens?

```
SELECT DISTINCT ?name ?height WHERE {
  SERVICE <{ontop}> {
    SELECT DISTINCT ?name ?height WHERE {
      ?building obe:hasPropertyUsage ?propertyUsage
          .
      ?propertyUsage a obe:School.
      ?propertyUsage obe:hasUsageLabel ?name.
      ?building bldg:lod0FootPrint ?footprintID;
                bldg:measuredHeight ?height.
    }
    ORDER BY DESC(?height)
    LIMIT 3
  }
}
```

Appendix C. Agent UML diagrams

See Figs. C.11 and C.12.

Appendix D. Data to ontology mapping tables

OpenStreetMap: <https://github.com/cambridge-cares/TheWorldAvatar/blob/main/Agents/OSMAgent/osmagent/src/main/resources/osm_tags.csv>

Municipal Land Plot: <https://github.com/cambridge-cares/TheWorldAvatar/blob/main/Agents/OSMAgent/osmagent/src/main/resources/dlm_landuse.csv>

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